

SUMMARY TABLE: CHEMISTRY GRADE 10

Sub-Strand 1.5: Periodicity — Teacher Reference (Pre-filled)

SUMMARY TABLE: CHEMISTRY GRADE 10	
Sub-Strand	Sub-Strand 1.5: Periodicity
Driving Question	DRIVING QUESTION: "Why do elements that sit next to each other in the periodic table — like sodium and magnesium, or chlorine and argon — behave so differently, and can we use an element's position in the table to predict exactly how it will behave before we ever put it in a test tube?"

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Something Strange at the Water Plant	Students watched sodium metal dropped into Lake Victoria water — it skimmed the surface, fizzed violently, ignited with an orange flame, and turned the water alkaline (pink with phenolphthalein). Magnesium in the same water barely fizzed. Aluminium showed no reaction. Argon in its sealed ampoule was completely inert.	Group I metals (Li, Na, K) show clear trends in physical properties going down the group: atomic radius increases, melting point decreases, and density generally increases. Reactivity with water also increases down the group ($\text{Li} < \text{Na} < \text{K}$). Going across Period 3 left to right, reactivity with cold water decreases sharply ($\text{Na} > \text{Mg} > \text{Al} > \text{Ar} \approx 0$). These patterns suggest that position in the periodic table is strongly connected to behaviour — but the class does not yet know WHY.	The class cannot yet fully explain why adjacent elements behave so differently. Current thinking is that it has something to do with what is inside the atom — specifically how electrons are arranged — but the mechanism (electron configuration, shielding, ionisation energy) has not yet been introduced. All unanswered questions are recorded on the Driving Question Board and will be addressed across subsequent lessons.
Lesson 2 Mapping the Group I Family	Lithium reacted slowly and steadily with cold water, floating on the surface and fizzing gently. Sodium reacted faster, skimming across the surface and fizzing vigorously — consistent with the Kisumu water plant demonstration. Potassium reacted most violently, producing a lilac flame and very rapid fizzing. All three produced a gas that burned with a squeaky pop when a lighted splint was held to it, confirming hydrogen production. All three produced alkaline solutions (pH 8 to 14),	All Group I metals have one electron in their outermost shell. Going down the group, atomic radius increases and additional electron shells increase the shielding effect, reducing the pull of the nucleus on the outer electron. This means the outer electron is more easily removed — lower ionisation energy — so the metal reacts more readily. The products of all three reactions are a metal hydroxide (LiOH , NaOH , KOH) and hydrogen gas (H_2). The alkaline solution results from the metal hydroxide dissolving	The dramatic difference in reactivity between sodium and potassium — despite both being in Group I — is explained by the increasing number of electron shells and increasing shielding as you go down the group, which progressively lowers ionisation energy. This begins to answer the driving question: position in the periodic table (specifically, which period an element occupies) determines how easily it loses its outer electron, and therefore how reactive it is. This also explains why the Kisumu water

	with potassium producing the highest pH reading.	in water.	officer stored sodium in argon gas — the sodium outer electron is easily lost even to trace moisture in air, so a completely inert atmosphere is required.
Lesson 3 Magnesium, Calcium and the Stubborn Metals	Students observed: (1) magnesium ribbon producing very slow or barely visible bubbles in cold water, with the solution turning faintly alkaline (universal indicator green-yellow, pH ~9–10 after several minutes); (2) magnesium burning vigorously in steam with a bright white flame, producing white MgO ash; (3) magnesium reacting rapidly with dilute HCl, producing vigorous bubbling and a warm test tube, with the gas giving a squeaky pop with a lighted splint; (4) calcium reacting noticeably faster than magnesium with cold water, producing steady bubbles and a distinctly alkaline solution (pH ~12). Students graphed: atomic radius increasing down Group II (Be → Ba), first ionisation energy decreasing down Group II, and melting point decreasing down Group II.	Students learned: (1) Group II metals are less reactive than Group I metals in the same period because they must lose TWO electrons rather than one, and their higher nuclear charge (more protons) makes those electrons harder to remove, resulting in a higher first ionisation energy; (2) reactivity within Group II increases DOWN the group because additional electron shells increase shielding, reducing the effective nuclear charge felt by outer electrons and lowering the ionisation energy; (3) the chemical reactions of magnesium with cold water, steam, dilute HCl and oxygen follow predictable patterns that can be expressed as balanced equations; (4) calcium's stronger reaction with cold water compared to magnesium is consistent with the down-group trend predicted by their model.	Students explained: Using the 'Two-Electron Tax' frame, magnesium is less reactive than sodium with cold water because Mg must lose two electrons (2,8,2) while Na loses only one (2,8,1); moreover, Mg has one extra proton, increasing the nuclear charge that holds the outer electrons more tightly, raising the first ionisation energy. Going from magnesium to calcium, reactivity increases because the extra electron shell in calcium (2,8,8,2) shields the outer electrons from the nucleus more effectively, lowering the ionisation energy despite the greater nuclear charge. This explains the observed trend: calcium reacted faster than magnesium with cold water, and both reacted far less dramatically than sodium — just as the water purification officer in Kisumu would predict if she switched from sodium to a Group II metal in her demonstration at Lake Victoria.
Lesson 4 The Halogen Family: From Gas to Solid	Learners observed: (1) physical properties of Cl₂ (yellow-green gas), Br₂ (orange-brown liquid) and I₂ (grey-black solid with violet vapour) from samples/data cards, noting the trend in colour and physical state down Group VII; (2) a strip of brightly coloured kanga fabric being bleached within 60 seconds by chlorine water, connecting to the Kisumu water treatment phenomenon; (3) iron wool forming brown iron(III) chloride in chlorine gas; (4) chlorine water turning the cyclohexane layer orange-brown when added to potassium bromide solution (Cl₂ displacing Br⁻); (5) bromine water turning the cyclohexane layer violet when added to potassium iodide solution (Br₂ displacing I⁻).	Learners learned that: (1) halogens exist in all three physical states at room temperature — Cl₂ (gas), Br₂ (liquid), I₂ (solid) — with colour deepening and volatility decreasing down the group; (2) chlorine reacts with water to form a bleaching solution (hypochlorous acid), explaining its use as a disinfectant in Lake Victoria water treatment; (3) reactivity in Group VII DECREASES down the group — the opposite of Group I — because atomic radius increases and electron affinity decreases, making it harder for larger atoms to attract an incoming electron; (4) displacement reactions (Cl₂ > Br₂ > I₂) reflect relative oxidising power and can be used to establish a reactivity series for halogens; (5) balanced equations for halogen displacement reactions require careful attention to stoichiometry and state	Learners explained that: (1) chlorine is the most reactive aqueous halogen because it has the smallest atomic radius and highest electron affinity among the three, allowing its nucleus to attract an incoming electron most strongly — this is why the Kisumu water officer uses Cl₂, not Br₂ or I₂, to treat Lake Victoria water; (2) the Group VII trend is the MIRROR IMAGE of the Group I trend — in Group I, more electron shells make it easier to LOSE an outer electron (lower ionisation energy → more reactive); in Group VII, more electron shells make it harder to GAIN an electron (lower electron affinity → less reactive); (3) displacement reactions prove relative oxidising power: a more reactive halogen (stronger oxidising agent) will displace a less reactive halide ion from solution; (4) the anchoring phenomenon's use of argon as an inert

		symbols.	shield (not yet fully explained) is now seen as the extreme end of a trend — as groups go right across the period, reactivity with electron gain becomes impossible for a full-shell noble gas, previewing the Lesson 5 across-period investigation.
Lesson 5 The Nobles and the Patterns of Eight	Students examined a data table of Group VIII elements (He, Ne, Ar, Kr) showing atomic radius, boiling point and first ionisation energy, and plotted four separate physical-property graphs across Period 3 (Na → Ar) — first ionisation energy, atomic radius, melting point and electronegativity — during a structured group graph-plotting relay, then displayed all four graphs as a 'Period 3 Properties Wall' for a class gallery walk.	Group VIII elements are chemically inert because their outer electron shells are completely full (He: 2 electrons; Ne, Ar, Kr: 8 electrons), making them extremely difficult to ionise; across Period 3, atomic radius generally decreases as nuclear charge increases while shielding remains nearly constant; first ionisation energy generally increases across Period 3 with characteristic dips at Al (3p sub-shell electron) and S (paired 3p electron repulsion); melting point peaks at Si due to its giant covalent structure; electronegativity increases Na → Cl with no value for Ar; Na, Mg and Al are malleable, ductile and electrically conductive, with conductivity increasing as the number of delocalised electrons per atom increases.	The water purification officer stored the reactive sodium in an argon atmosphere because argon's completely full outer shell (2,8,8) gives it a first ionisation energy of 1521 kJ/mol — more than three times that of sodium (496 kJ/mol) — meaning its electrons cannot be removed or shared under any ordinary chemical conditions, making argon perfectly inert and safe as a shielding gas; the dramatic reactivity difference between sodium and its Period 3 neighbours (Mg, Al) that students observed in the anchoring phenomenon can now be quantitatively explained by the steep rise in first ionisation energy from left to right across the period — each additional proton pulling the outer electrons more tightly — while the anomalies in that trend reveal the deeper sub-shell structure of the atom.
Lesson 6 Period 3 Reacts: Metals, Non-Metals and the Chlorine Test	Sodium reacted violently with cold water producing a colourless gas that burned with a pop, leaving an alkaline solution; magnesium reacted very slowly with cold water but vigorously with dilute HCl; aluminium showed no visible reaction with cold water; magnesium and sulfur burned brightly in oxygen; the vigour of reaction decreased from left to right across period 3.	The reactivity of period 3 elements with water and acids decreases across the period because nuclear charge increases while shielding remains nearly constant, generally raising ionisation energy (Al first ionisation energy dips below Mg, but removing all three of its outer electrons costs far more energy) and making it harder to remove outer electrons; period 3 metals form basic oxides and non-metals form acidic oxides; the products of reactions with water and HCl can be predicted from electron configuration.	Sodium reacts explosively with Lake Victoria water because it has only one outer electron held by 11 protons with 2 full inner shells providing moderate shielding — its ionisation energy is only 496 kJ per mole and the electron is given up almost instantaneously to water molecules; magnesium has 12 protons with the same inner shielding, so its two outer electrons experience stronger effective nuclear charge and are harder to remove — reactivity is far lower; aluminium and silicon have even higher effective nuclear charges and more outer electrons to remove (Al first ionisation energy, 577 kJ per mole, dips below Mg at 738 because its single 3p electron is easier to remove, but the total energy to remove all three outer electrons is much higher, and Al is also protected by its oxide layer), so they do not react with cold water at all; the anchoring phenomenon is

			now fully explained by the left-to-right increase in effective nuclear charge across period 3.
Lesson 7 What Are These Elements Good For? — Uses, Applications & Final Explanation	Students observed (across the unit): sodium metal exploding violently in Lake Victoria water, skimming and igniting with an orange flame; magnesium barely fizzing in cold water; aluminium showing no reaction with cold water; and argon being completely inert. They observed a clear left-to-right decrease in reactivity across Period 3 and a clear top-to-bottom increase in reactivity down Group I (lithium < sodium < potassium). In this lesson, students observed expert group presentations revealing that these same reactivity differences directly determine real-world uses — chlorine's reactivity makes it a water disinfectant, argon's inertness makes it a welding shield, sodium's low ionisation energy underlies both its explosive water reaction and its use in sodium vapour street lamps.	Students learned that an element's position in the periodic table — specifically its group and period number — determines its electron configuration, which in turn determines every physical and chemical property observed. Going down Group I (Li → Na → K), increasing electron shells increase shielding, reducing the effective nuclear charge felt by the outer electron, lowering ionisation energy and increasing reactivity with water. Going across Period 3 (Na → Mg → Al), increasing nuclear charge with the same shielding effect generally increases ionisation energy (with a dip at Al, whose 3p electron is easier to remove, although removing all three Al outer electrons costs more energy) and decreases reactivity with cold water. Group VIII elements have full outer shells (noble gas configuration), giving them zero tendency to gain or lose electrons, making them completely inert. These principles explain every Kenyan application studied: chlorine's 7 valence electrons drive its oxidising disinfectant power; argon's full shell makes it the ideal inert welding atmosphere; calcium's 2 valence electrons and moderate reactivity make it suitable for construction chemistry in cement.	Students can now explain the anchoring phenomenon in full: The water purification officer's sodium explodes in Lake Victoria water because sodium's electron configuration [2,8,1] — with a single outer electron shielded by 10 inner electrons — gives sodium an exceptionally low first ionisation energy (496 kJ/mol). This outer electron is so easily removed that sodium reacts spontaneously and violently with water molecules, releasing hydrogen gas (which ignites) and forming strongly alkaline NaOH. Magnesium [2,8,2] has two outer electrons held by a nuclear charge of +12 but shielded by only 10 inner electrons — a much higher effective nuclear charge — so its ionisation energy is higher and it reacts only slowly with cold water. Argon [2,8,8] has a completely full outer shell with no tendency whatsoever to donate, accept, or share electrons, making it perfectly inert — exactly why the officer uses it to shield sodium during transport. Potassium [2,8,8,1] has its outer electron in a fourth shell, further from the nucleus and shielded by 18 inner electrons, so its ionisation energy is even lower than sodium's, making it explode more violently. Position in the periodic table therefore allows chemists — and Kenyan water treatment engineers, construction workers, and medical professionals — to predict exactly how an element will behave before it ever enters a test tube.

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